

Chapter 1

Arbitrage-Free Modelling of Effective Interest Rates

In this first chapter we provide a general framework for modelling of effective forward interest rates such as forward Libor and swap rates. Throughout the whole book Libor may stand for LIBOR (London Inter Bank Offer Rate), EurIBOR (European Inter Bank Offer Rate), or an inter bank rate in some other currency environment.

1.1 Elements of Arbitrage Theory and Derivative Pricing

We here review some basic principles of arbitrage theory and derivative pricing. The results are formulated for price systems which are governed by Itô processes, processes which can be represented by stochastic Itô-integrals with respect to Brownian motion. A summary of the involved stochastic calculus and other relevant probabilistic tools is given in Appendix A.1.

1.1.1 Arbitrage-free Systems, Self-Financing Trading Strategies, Complete Markets

Let $(\Omega, \mathcal{F}, P)$ be a probability space with a filtration $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T_\infty < \infty}$ generated by an $\mathbb{R}^m$ -valued Brownian motion W which satisfies the “usual condition”. The filtration is thus restricted to a fixed finite interval. Our starting point is an n -dimensional process $B = (B(t))_{0 \leq t \leq T_\infty}$ of tradable securities given by the SDE

$$\frac{dB_i(t)}{B_i(t)} = \mu_i(t) dt + \sum_{j=1}^m \sigma_{ij}(t) dW_j(t), \quad B_i(0) > 0, \quad i = 1, \dots, n. \quad (1.1)$$

For technical reasons, the coefficient processes μ and σ are assumed to be $\mathbb{F}$ -predictable (continuous from the left is enough), and such that $\int_0^{T_\infty} |\mu_i(u)| du < \infty$ P -a.s. The $\mathbb{R}^m$ -valued process $\sigma_i = (\sigma_{ij})_{j=1, \dots, m}$ is assumed to be W -integrable for each i . This ensures in particular that (1.1) has a unique strong

solution with values in $\mathbb{R}_{++}^n := \{x \in \mathbb{R}^n \mid x_i > 0\}$ for $i = 1, \dots, n$. In our context of modelling interest rate dynamics we think of B as describing the price evolution of n risky bonds traded in a financial market where prices are expressed in some fixed currency unit.

DEFINITION 1.1 *The price system (market) B in (1.1) is said to be **Arbitrage-free (AF)** if there exists an adapted process ξ on $(\Omega, \mathcal{F}, P)$ with $\xi > 0$ and $\xi_0 = 1$, such that ξB_i are martingales for all $1 \leq i \leq n$. The process ξ is called a *state price deflator*.*

Note that in (1.1) the *state price deflator* makes deflated prices martingales in the *actual measure*. See also Duffie (2001). From the fact that ξ is adapted to the Brownian filtration $\mathcal{F}$ and the fact that the B_i are positive Itô processes, it is not difficult to show by the martingale representation theorem that ξ satisfies also an Itô SDE,

$$\frac{d\xi(t)}{\xi(t)} = -r(t) dt - \sum_{j=1}^m \lambda_j(t) dW_j(t), \quad \xi(0) = 1, \quad (1.2)$$

for a certain predictable scalar process r and vector process λ in $\mathbb{R}^m$. We will refer to r as the (virtual) *short rate* and λ as the *market price of risk* process.

DEFINITION 1.2 *A trading strategy in the above market is an $\mathbb{R}^n$ -valued $\mathbb{F}$ -predictable B -integrable process φ . If φ with corresponding value process*

$$V^\varphi := \sum_{i=1}^n \varphi_i B_i =: \varphi^\top B$$

satisfies the self-financing condition

$$V^\varphi(t) = V^\varphi(0) + \int_0^t \varphi(u)^\top dB(u), \quad 0 \leq t \leq T_\infty, \quad (1.3)$$

*we speak of a **Self-Financing Trading Strategy (SFTS)** in the market B .*

It should be noted that in a more general setup one usually requires the processes ξB_i in Definition 1.1 to be local martingales rather than true martingales. For the present exposition, however, we want to avoid technical details related to the issue of local martingales versus true martingales, and therefore we assume in Definition 1.1 more strongly the martingale property. For similar reasons we restrict the class of trading strategies in an arbitrage-free system B to the class of *admissible strategies* in the system B .

DEFINITION 1.3 For an arbitrage-free system B with price deflator ξ we call a trading strategy φ **admissible** if the process $t \rightarrow \int_0^t \varphi^\top d(\xi B)$ (which is a local martingale by Theorem A.1) is a true martingale on $[0, T_\infty]$.

Without going into details, we here just note that a local martingale is a true martingale under certain extra integrability conditions, e.g., Novikov's conditions. The economical relevance of an arbitrage-free price system B is that, within this system, there is no possibility for a "free lunch" in the following sense.

THEOREM 1.1

Assume B in (1.1) is arbitrage-free with price deflator ξ , and $T \leq T_\infty$. If φ is an admissible self-financing trading strategy with initial value $V^\varphi(0) = \varphi(0)^\top B(0) = 0$ and $V^\varphi(T) \geq 0$ a.s., then $V^\varphi(T) = 0$ a.s.

The proof of this theorem is based on the following lemma

LEMMA 1.1

Let φ be a self-financing trading strategy in the system B , i.e., (1.3) holds. Then, for any positive continuous adapted semimartingale ξ such that φ is ξ -integrable we have

$$\varphi(T)^\top \xi(T) B(T) = \varphi(0)^\top \xi(0) B(0) + \int_0^T \varphi(u)^\top d(\xi(u) B(u)), \quad (1.4)$$

so φ is also a self-financing trading strategy in the system ξB .

PROOF By the SFTS property $d(\varphi^\top B) = \varphi^\top dB$ the value process $\varphi^\top B$ is a continuous semimartingale. Then, by applying two times the product rule (A.11) for continuous semimartingales and again the self-financing property, it follows that, in condensed notation,

$$\begin{aligned} d(\varphi^\top(\xi B)) &= \xi d(\varphi^\top B) + \varphi^\top B d\xi + d(\varphi^\top B) \cdot d\xi \\ &= \xi \varphi^\top dB + \varphi^\top B d\xi + (\varphi^\top dB) \cdot d\xi \\ &= \varphi^\top (\xi dB + B d\xi + dB \cdot d\xi) \\ &= \varphi^\top d(\xi B), \end{aligned}$$

where $dX \cdot dY := d\langle X, Y \rangle$, with $\langle X, Y \rangle$ being the covariation process of X and Y . $\square$

PROOF (Theorem 1.1) Since B is arbitrage-free there exists by definition a price deflator $\xi > 0$ such that the ξB_i are martingales, and ξ is continuous by (1.2). By using the admissibility of the SFTS for the system (ξ, B) ,

Lemma 1.1, and $\varphi(0)^\top B(0) = 0$, it then follows that $E\varphi(T)^\top \xi(T)B(T) = 0$ by taking the expectation of (1.4). Hence, $\varphi(T)^\top \xi(T)B(T) = 0$ a.s. and so $\varphi(T)^\top B(T) = 0$, a.s. since $\xi > 0$. $\square$

For an arbitrage-free system the dynamics of asset prices (1.1) and price deflator (1.2) can be written in integrated form,

$$\begin{aligned} B_i &= B_i(0) \exp \left[\int_0^t (\mu_i - \frac{1}{2}|\sigma_i|^2) ds + \int_0^t \sigma_i^\top dW \right], \\ \xi &= \exp \left[\int_0^t (-r - \frac{1}{2}|\lambda|^2) ds - \int_0^t \lambda^\top dW \right]. \end{aligned} \quad (1.5)$$

Equations (1.5) are easily verified by applying Itô's lemma (A.10) to them, which yields (1.1) and (1.2) again. From the representation (1.5) it follows that

$$\xi B_i = B_i(0) \exp \left[\int_0^t (\mu_i - r - \frac{1}{2}|\sigma_i|^2 - \frac{1}{2}|\lambda|^2) ds + \int_0^t (\sigma_i - \lambda)^\top dW \right],$$

and so, since ξB_i are martingales for every i ,

$$-\frac{1}{2}|\sigma_i - \lambda|^2 = \mu_i - r - \frac{1}{2}|\sigma_i|^2 - \frac{1}{2}|\lambda|^2,$$

or equivalently,

$$\mu_i = r + \sigma_i^\top \lambda, \quad \text{for } i = 1, \dots, n. \quad (1.6)$$

We conclude that for an arbitrage-free price system (1.1) the dynamics of a price deflator ξ can be written in the form (1.2), where r and λ satisfy (1.6). Conversely, if there exist for a price system (1.1) predictable processes r and λ such that (1.6) holds, and the solution ξ of (1.2) is such that the processes ξB_i satisfy a certain integrability condition (e.g., Novikov's condition), then the system is arbitrage-free.

We next introduce the concept of completeness which essentially comes down to perfect replicability of all measurable claims.

DEFINITION 1.4 *Let the price system B in (1.1) be arbitrage-free. Then the system is said to be **complete** if the price deflator $\xi > 0$, with ξB_i being martingales for all i and $\xi_0 = 1$, is **unique**.*

If we impose a full-rank restriction on the volatility structure of system (1.1), we have the following proposition by Definition 1.4, no-arbitrage condition (1.6), and some linear algebra.

PROPOSITION 1.1

Suppose for each (t, ω) the $n \times m$ matrix σ , defined by $\sigma[i, k] := \sigma_i[k]$ has full rank m , hence $m \leq n$. Then with $\mathbf{1} := [1, \dots, 1]^\top \in \mathbb{R}^n$ we have

- For $m = n$, the market is arbitrage-free but incomplete.
- If $m = n - 1$ and
 - $\mathbf{1} \notin \text{range}(\sigma)$, the market is arbitrage-free and complete.
 - $\mathbf{1} \in \text{range}(\sigma)$,
 - * the market is arbitrage-free but incomplete if $\mu \in \text{span}\{\mathbf{1}, \sigma\}$.
 - * there is arbitrage if $\mu \notin \text{span}\{\mathbf{1}, \sigma\}$.
- If $m < n - 1$ and
 - $\mu \in \text{span}\{\mathbf{1}, \sigma\}$, the market is arbitrage-free and
 - * complete if $\mathbf{1} \notin \text{range}(\sigma)$.
 - * incomplete if $\mathbf{1} \in \text{range}(\sigma)$.
 - $\mu \notin \text{span}\{\mathbf{1}, \sigma\}$, there is arbitrage.

The next theorem and Theorem 1.1 are essentially due to Harrison & Pliska (1981, 1983) and Delbaen and Schachermayer (1994).

THEOREM 1.2

If the arbitrage-free price system B in (1.1) is complete, then for any T , $0 \leq T \leq T_\infty$, and any random variable $C_T \in \mathcal{F}_T$ (an $\mathcal{F}_T$ -claim), there exists an SFTS φ and an initial investment $V^\varphi(0)$ such that

$$C_T = V^\varphi(T) = V^\varphi(0) + \int_0^T \varphi(t)^\top dB(t).$$

The value of $V^\varphi(0)$ is considered as the fair price of the claim.

PROOF We will sketch a proof of Theorem 1.2 under assumption of the full-rank condition in Proposition 1.1. For a fixed but arbitrary claim the martingale $E^{\mathcal{F}_t}(\xi(T)C_T)$ can be represented by the martingale representation theorem as

$$E^{\mathcal{F}_t}(\xi(T)C_T) = E^{\mathcal{F}_0}(\xi(T)C_T) + \int_0^t \xi \alpha^\top dW, \quad (1.7)$$

for some predictable process α in $\mathbb{R}^m$. For any predictable process φ we have

$$\int_0^t \varphi^\top d(\xi B) = \int_0^t \sum_i \varphi_i \xi B_i (\sigma_i - \lambda)^\top dW. \quad (1.8)$$

By Proposition 1.1, completeness implies $\mathbf{1} \notin \text{range}(\sigma)$ and $\text{rank}(\sigma) < n$. So, in particular, the $n \times m$ matrix $\sigma - \mathbf{1}\lambda^\top$ has full rank m also and thus the

$m \times m$ matrix $(\sigma - \mathbf{1}\lambda^\top)^\top(\sigma - \mathbf{1}\lambda^\top)$ is invertible. Therefore, the solution space of the linear system $y^\top(\sigma - \mathbf{1}\lambda^\top) = \alpha^\top$ can be written as

$$y = (\sigma - \mathbf{1}\lambda^\top) \left((\sigma - \mathbf{1}\lambda^\top)^\top (\sigma - \mathbf{1}\lambda^\top) \right)^{-1} \alpha + \mathcal{N},$$

with $\mathcal{N} = \{v : v^\top(\sigma - \mathbf{1}\lambda^\top) = 0\}$. Note that

$$\begin{aligned} \mathcal{N}_0 &:= \{v : v^\top(\sigma - \mathbf{1}\lambda^\top) = 0 \wedge v^\top \mathbf{1} = 0\} \\ &= \{v : v^\top \sigma = 0 \wedge v^\top \mathbf{1} = 0\} \\ &= \text{span}^\perp \{\text{range}(\sigma), \mathbf{1}\} \subset \mathcal{N}, \end{aligned}$$

where $0 \leq \dim \mathcal{N}_0 = n - m - 1 < \dim \mathcal{N} = n - m$. It thus follows that there exists a predictable process η in $\mathbb{R}^n$ such that $\sum_i \eta_i(\sigma_i - \lambda) = \alpha$ with $\sum_i \eta_i(t) = \eta(t)^\top \mathbf{1} = \xi^{-1}(t)E^{\mathcal{F}_t}(\xi(T)C_T)$, namely,

$$\eta = (\sigma - \mathbf{1}\lambda^\top) \left((\sigma - \mathbf{1}\lambda^\top)^\top (\sigma - \mathbf{1}\lambda^\top) \right)^{-1} \alpha + v,$$

with

$$v^\top \mathbf{1} = \xi^{-1}(t)E^{\mathcal{F}_t}(\xi(T)C_T) - \mathbf{1}^\top(\sigma - \mathbf{1}\lambda^\top) \left((\sigma - \mathbf{1}\lambda^\top)^\top (\sigma - \mathbf{1}\lambda^\top) \right)^{-1} \alpha.$$

Moreover, if $m = n - 1$ the solution process η is unique.

Let us now consider the trading strategy $\varphi_i = \eta_i/B_i$ for the system B . We will show that φ is self-financing and that $\varphi^\top(T)B(T) = C_T$, hence the corresponding self-financing portfolio replicates the claim. From (1.7) and (1.8) we obtain

$$\begin{aligned} \int_0^t \varphi^\top d(\xi B) &= \int_0^t \xi \alpha^\top dW \\ &= E^{\mathcal{F}_t}(\xi(T)C_T) - E^{\mathcal{F}_0}(\xi(T)C_T) \\ &= \xi(t)\varphi^\top(t)B(t) - \xi(0)\varphi^\top(0)B(0); \end{aligned}$$

hence the pair $(\varphi, \xi B)$ satisfies the self-financing condition (1.3). Then, applying Lemma 1.1 with ξ replaced by ξ^{-1} and B replaced by ξB yields that the pair (φ, B) satisfies the self-financing condition (1.3) as well. So (φ, B) is a self-financing portfolio and

$$\begin{aligned} \varphi(t)^\top B(t) &= \xi^{-1}(t)E^{\mathcal{F}_t}(\xi(T)C_T) \\ &= \xi^{-1}(0)E^{\mathcal{F}_0}(\xi(T)C_T) + \int_0^t \varphi^\top dB, \end{aligned}$$

where in particular $\varphi(T)^\top B(T) = C_T$ and $\xi^{-1}(0)E^{\mathcal{F}_0}(\xi(T)C_T)$ is considered to be the fair price of the claim at time $t = 0$. $\square$

REMARK 1.1 In the case where the system (1.1) constitutes a complete market (see Proposition 1.1), it can be shown along the same lines as in the

proof of Theorem 1.2 (e.g., see also Reiß, Schoenmakers, Schweizer 2001) that there exists a portfolio β_t generated by some self-financing trading strategy, such that $\beta(t) = \beta(0) \exp(\int_0^t r(s)ds)$; hence β plays the role of a savings account. However, following Jamshidian (1997), we explain in the next section that assuming the existence of such a savings account is not necessary for pricing and hedging of Libor derivative products. $\square$

1.1.2 Derivative Claim Pricing in Different Measures

As we saw in Section 1.1.1 (Theorem 1.2), in a complete market any measurable claim C_T can be replicated by a self-financing portfolio and the initial value of this portfolio is the fair price of the claim. If the market is arbitrage-free but incomplete, the collection of replicable claims is generally smaller but not empty. For instance, any claim of type $C_T = \sum_i^n \alpha_i B_i(T)$ with constant α 's is trivially replicable.

Let us suppose C_T is a replicable claim in an arbitrage-free system (1.1) and let φ be a self-financing trading strategy which replicates the claim, i.e.,

$$C_T = \varphi(T)^\top B(T) = \varphi(0)^\top B(0) + \int_0^T \varphi(t)^\top dB(t).$$

Since (1.1) is arbitrage-free there exists a price deflator ξ such that ξB_i are martingales. By Lemma 1.1 we have

$$\xi(T)C_T = \xi(T)\varphi(T)^\top B(T) = \xi(0)\varphi(0)^\top B(0) + \int_0^T \varphi(s)^\top d(\xi(s)B(s)),$$

and by the martingale property of the process $\int_0^t \varphi^\top d(\xi B)$ we get for $t \leq T$,

$$E^{\mathcal{F}_t}(\xi(T)C_T) = \xi(t)\varphi(t)^\top B(t) = \xi(0)\varphi(0)^\top B(0) + \int_0^t \varphi(s)^\top d(\xi(s)B(s)),$$

hence,

$$\varphi(t)^\top B(t) = \frac{1}{\xi(t)} E^{\mathcal{F}_t}(\xi(T)C_T). \quad (1.9)$$

From (1.9) we conclude the following. On the one hand, in an incomplete market where the price deflator process ξ is not unique, the right-hand side of (1.9) does not depend on ξ . On the other hand, if $\tilde{\varphi}$ is another SFTS which replicates C_T , we have $\varphi(t)^\top B(t) = \tilde{\varphi}(t)^\top B(t)$, i.e., the two SFTS's have always the same price. Thus, the time $t < T$ value of the claim C_T is properly defined by

$$C_t := \frac{1}{\xi(t)} E^{\mathcal{F}_t}(\xi(T)C_T). \quad (1.10)$$

As a result, in a complete market where any $\mathcal{F}_T$ -measurable claim C_T can be hedged by an SFTS (Theorem 1.2), the price C_t of this claim at a prior time $t < T$ is given by (1.10).

In (1.10) the price of a derivative claim is represented with respect to the objective or “real world” measure. We now introduce the notion of *numeraire measures* and will give representations for the claim price C_t in different numeraire measures by using numeraire transformations.

DEFINITION 1.5 *Let the system (1.1) be arbitrage-free and ξ be a price deflator such that ξB_i are martingales. Let further $A > 0$ be a positive adapted process such that ξA is a martingale. The A -numeraire measure P_A is then defined via the Radon-Nykodym derivative*

$$\frac{dP_A}{dP} := \frac{\xi(T_\infty)A(T_\infty)}{A(0)}, \quad \text{by}$$

$$E_A X := \int X dP_A := \int X \frac{dP_A}{dP} dP = E \left(X \frac{\xi(T_\infty)A(T_\infty)}{A(0)} \right),$$

for any $\mathcal{F}$ -measurable random variable X . Moreover, if X_t is $\mathcal{F}_t$ -measurable, we have by the martingale property of ξA ,

$$E_A X_t = E E^{\mathcal{F}_t} \left(X_t \frac{\xi(T_\infty)A(T_\infty)}{A(0)} \right) = E \left(X_t \frac{\xi(t)A(t)}{A(0)} \right) = E (X_t M_A(t)),$$

where the Radon-Nykodym process

$$M_A(t) := E^{\mathcal{F}_t} \frac{dP_A}{dP} = \frac{\xi(t)A(t)}{A(0)}$$

is a martingale associated with the numeraire A .

The following lemma is quite useful and easy to prove.

LEMMA 1.2

If ξA and ξX are martingales then X/A is a P_A martingale.

PROOF By a well-known rule for the conditional expectation of a Radon-Nykodym transformed measure, we have for $0 \leq t \leq t+s \leq T_\infty$,

$$\begin{aligned} E_A^{\mathcal{F}_t} \frac{X(t+s)}{A(t+s)} &= \frac{E^{\mathcal{F}_t} \left(\frac{dP_A}{dP} \frac{X(t+s)}{A(t+s)} \right)}{E^{\mathcal{F}_t} \frac{dP_A}{dP}} = \frac{E^{\mathcal{F}_t} (M_A(t+s) X(t+s) / A(t+s))}{M_A(t)} \\ &= \frac{E^{\mathcal{F}_t} (\xi(t+s) X(t+s) / A(0))}{\xi(t) A(t) / A(0)} = \frac{X(t)}{A(t)}. \end{aligned}$$

□

We now have a representation for the option price (1.10) in terms of the numeraires A in Definition 1.5, due to the following proposition.

PROPOSITION 1.2

As in Definition 1.5, let $A > 0$ and ξA be a martingale. Let further C_T be an option ($\mathcal{F}_T$ -claim) which can be hedged by an SFTS. Then, we have

$$\begin{aligned} C_t &= \frac{1}{\xi(t)} E^{\mathcal{F}_t} (\xi(T) C_T) \\ &= A(t) E_A^{\mathcal{F}_t} \frac{C_T}{A(T)}. \end{aligned} \quad (1.11)$$

PROOF Since ξC and ξA are martingales it follows from lemma (1.2) that C/A is a P_A martingale, hence $C_t/A(t) = E_A^{\mathcal{F}_t} (C_T/A(T))$. $\square$

Proposition 1.2 is particularly useful in the case where A is one of the assets in the system (1.1), i.e., $A = B_i$ for some i . Moreover, if B_i is a T_i -maturity zero coupon bond with $B_i(T) = 1$, we get simply $C_t = B_i(t) E_{B_i}^{\mathcal{F}_t} C_T$.

1.2 Modelling of Effective Forward Rates

In this section we will derive the dynamics of forward Libor and swap rate processes defined with respect to a fixed system of tenor dates. Since both Libor and swap rates can be expressed in terms of zero-coupon bonds, it is natural to derive their respective dynamics from an arbitrage-free system (1.1) consisting of (zero) bonds B_i . The Libor rate process will be studied in Section 1.2.1 and the swap rate dynamics in Section 1.2.2.

1.2.1 Libor Rate Processes and Measures

Let us consider a fixed sequence of tenor dates $0 < T_1 < T_2 < \dots < T_n < T_\infty$, called a tenor structure, together with a sequence of so called day-count fractions $\delta_i := T_{i+1} - T_i$, $i = 1, \dots, n-1$. With respect to this tenor structure we consider an arbitrage-free system (1.1) of zero-bond processes B_i , for $i = 1, \dots, n$, where each B_i lives on the interval $[0, T_i]$ and ends up with its face value $B_i(T_i) = 1$. With respect to this bond system we deduce a system of forward rates, called Libor rates, which are defined by

$$L_i(t) := \frac{1}{\delta_i} \left(\frac{B_i(t)}{B_{i+1}(t)} - 1 \right), \quad 0 \leq t \leq T_i, \quad 1 \leq i \leq n-1. \quad (1.12)$$

Note that L_i in (1.12) is the annualized effective forward rate to be contracted for at the (present) date t , for a loan over a forward period $[T_i, T_{i+1}]$. Based on this rate one has to pay at T_{i+1} an interest amount of $\$ \delta_i L_i(T_i)$ on a \$1 notional.

We will now derive the dynamics of the Libor process L when the bond dynamics is given by the system (1.1) under the no-arbitrage condition (1.6). By applying Itô via Lemma A.2 to the Libor definition (1.12), using (1.6) and (1.12) again, we obtain straightforwardly

$$\begin{aligned} dL_i &= \delta_i^{-1}(1 + \delta_i L_i) \left((\mu_i - \mu_{i+1} - \sigma_{i+1}^\top (\sigma_i - \sigma_{i+1})) dt + (\sigma_i - \sigma_{i+1})^\top dW \right) \\ &= \delta_i^{-1}(1 + \delta_i L_i) (\sigma_i - \sigma_{i+1})^\top (dW + (\lambda - \sigma_{i+1}) dt), \quad 1 \leq i < n. \end{aligned}$$

By introduction of the Libor volatility processes $\gamma_i \in \mathbb{R}^m$,

$$L_i \gamma_i := \delta_i^{-1}(1 + \delta_i L_i) (\sigma_i - \sigma_{i+1}), \quad (1.13)$$

and the drifted Brownian motions

$$dW^{(j)} := dW + (\lambda - \sigma_j) dt, \quad 1 \leq j \leq n, \quad (1.14)$$

we may write for $1 \leq i < n$,

$$dL_i = L_i \gamma_i^\top dW^{(i+1)} \quad (1.15)$$

$$= - \sum_{j=i+1}^{n-1} \frac{\delta_j L_i L_j}{1 + \delta_j L_j} \gamma_i^\top \gamma_j dt + L_i \gamma_i^\top dW^{(n)}, \quad 0 \leq t \leq T_i. \quad (1.16)$$

From Definition 1.12 and Lemma 1.2 it follows that L_i is a martingale with respect to the measure $P_{B_{i+1}}$. From a general martingale representation theorem it then follows that $(W^{(i+1)}(t))_{0 \leq t \leq T_i}$ in (1.15) is standard Brownian motion under $P_{B_{i+1}}$, for $1 \leq i < n$. Similarly, since again by Lemma 1.2 $B_{i+1}/B_i = (1 + \delta_i L_i)^{-1}$ is a martingale under P_{B_i} , we can show that $(W^{(i)}(t))_{0 \leq t \leq T_i}$ is standard Brownian motion under P_{B_i} , for $1 \leq i < n$. By combining, we obtain the following corollary.

COROLLARY 1.1

For $j = 1, \dots, n$ the process

$$W^{(j)}(t) = W(t) + \int_0^t (\lambda - \sigma_j) ds, \quad 0 \leq t \leq T_j \wedge T_{n-1},$$

is an m -dimensional Brownian motion under the measures P_{B_j} .

REMARK 1.2 For each particular i the zero bond B_i terminates at T_i and so, after passing a maturity T_i , the number of “living bonds” in the bond system (1.1) decreases with one. As a consequence, the arbitrage theory in Section 1.1 should be considered on the time interval $[0, T_1]$ with respect to the full system $B_1, \dots, B_n$, then on the time interval $(T_1, T_2]$ with respect to $B_2, \dots, B_n$, and so on. In order to meet the full-rank condition in Proposition 1.1 in each such interval, it is natural to assume at most n Brownian motions, hence $m \leq n$, and $\sigma[i, k] = \sigma_i[k] = 0$, for $i - k > n - m$. $\square$

From now on we will use for the numeraire measure P_{B_j} the shorter notation P_j and for the related expectation we will write E_j . The Libor dynamics (1.16) is derived from a given arbitrage-free zero bond system where the volatility process γ can be expressed explicitly in the bond process via (1.13) and (1.12). Now the following question arises. Let us take some Libor volatility process γ . Does there exist an arbitrage-free zero-bond process (1.1) such that (1.12) holds? The following proposition gives a partially confirmative answer sufficient for our purposes (see Jamshidian (1997), Th. 5.3 and 7.1, for more details).

PROPOSITION 1.3

Existence of the Libor process *Suppose we are given a bounded and locally Lipschitz volatility structure of type $\gamma_i(t, L)$, $1 \leq i < n$. Then there exists an arbitrage-free system B of bond prices (1.1) satisfying $B_i(T_i) = 1$ and a price deflator ξ (i.e., ξB_i are P -martingales for $1 \leq i \leq n$), such that (1.16) with $\gamma_i = \gamma_i(t, L)$ has a unique solution for which (1.12) holds.*

Let us now start with a given bounded and locally Lipschitz volatility function $\mathbb{R} \times \mathbb{R}^{n-1} \ni (t, L) \rightarrow \gamma(t, L) \in \mathbb{R}^{(n-1) \times m}$. Consider the corresponding Libor process $L(t)$ as a solution of SDE (1.16), where $W^{(n)}$ is an m -dimensional Brownian motion with respect to P_n , starting at some $L(0) \in \mathbb{R}_+^{n-1}$. In fact, economically only the distribution of the Libor process is relevant. For this reason we assume without loss of generality that we have at most $n-1$ Brownian motions in (1.16), i.e., $m \leq n-1$, and that $\gamma(t, L)$ has full rank m . Indeed, if $m \geq n$ it would be possible to generate by (1.16) a Libor process with the same distribution, using another full rank volatility function with $m \leq n-1$. Due to Proposition 1.3, there exist at the background an arbitrage-free zero bond system (1.1) which satisfies $B_i(T_i) = 1$, $1 \leq i \leq n$, and (1.12). As a consequence, all ratios B_i/B_j are measurable with respect to L , hence the filtration generated by the m (at most $n-1$) Brownian motions. Moreover, in this setup the following basic result, essentially due to Jamshidian (1997) Th. 5.2, applies.

THEOREM 1.3

Hedging Libor derivative claims *Consider an arbitrage-free bond system (1.1) and the corresponding Libor process (1.12) with dynamics given by (1.16). Suppose that $W^{(n)}$ is an m -dimensional Brownian motion with $m < n$ and the volatility process γ has constant rank m and is predictable with respect to $\mathcal{J}$, the filtration generated by $W^{(n)}$. Then, any claim C_T such that $C_T/B_n(T)$ is measurable with respect to $\mathcal{J}_T$ can be priced and hedged by a self-financing trading strategy φ in the bond system B . The arbitrage-free price of*

the claim is given by (see also (1.11))

$$C_t := \varphi(t)^\top B(t) = B_n(t) E_n^{\mathcal{J}_t} \frac{C_T}{B_n(T)}. \quad (1.17)$$

PROOF We will sketch the proof which is similar to the proof of Theorem 1.2. By the martingale representation theorem there exists a $\mathcal{J}$ -predictable process $\beta \in \mathbb{R}^m$ such that

$$\frac{C_t}{B_n(t)} = E_n^{\mathcal{J}_t} \frac{C_T}{B_n(T)} = E^{\mathcal{J}_0} \frac{C_T}{B_n(T)} + \int_0^t \beta^\top dW^{(n)}.$$

Since L is $\mathcal{J}$ -adapted, the system $(B/B_n) := (B_i/B_n)_{1 \leq i \leq n} \in \mathbb{R}^{n-1}$ is a $\mathcal{J}$ -adapted martingale in P_n , and so $d(B/B_n) = \nu dW^{(n)}$ for some predictable process $\nu \in \mathbb{R}^{(n-1) \times m}$. It is not difficult to see that since γ has full rank, ν has full rank also, and so we may consider $\theta := \nu(\nu^\top \nu)^{-1} \beta \in \mathbb{R}^{n-1}$ yielding

$$\theta^\top d(B/B_n) = \beta^\top (\nu^\top \nu)^{-1} \nu^\top \nu dW^{(n)} = \beta^\top dW^{(n)} = d(C/B_n).$$

By taking $\varphi_i := \theta_i$, for $1 \leq i < n$ and $\varphi_n := C/B_n - \theta^\top (B/B_n)$, we thus have

$$d((\varphi^\top B)/B_n) = d(C/B_n) = \theta^\top d(B/B_n) = \sum_{i=1}^n \varphi_i d(B_i/B_n).$$

Hence, by Lemma 1.1 it follows that the pair (φ, B) constitutes a self-financing portfolio with

$$C_t = B_n(t) E_n^{\mathcal{J}_t} \frac{C_T}{B_n(T)} = \varphi(t)^\top B(t) = \varphi(0)^\top B(0) + \int_0^t \varphi^\top dB.$$

□

Note that in Theorem 1.3 completeness of the Bond system (1.1) is not required. As we will see, Theorem 1.3 applies to almost all Libor derivative claims in practice.

From (1.13) and Corollary 1.1 we easily derive for any i , $1 \leq k \leq n$,

$$dW^{(n)} = dW^{(k)} + \sum_{j=k}^{n-1} \frac{\delta_j L_j \gamma_j}{1 + \delta_j L_j} dt, \quad 0 \leq t \leq T_i \wedge T_{n-1}, \quad (1.18)$$

where an empty sum is defined to be 0. Thus, by inserting (1.18) in (1.16) we obtain the dynamics of L_i in any bond measure P_k , $1 \leq k \leq n$,

$$\begin{aligned} dL_i = & - \sum_{j=i+1}^{n-1} \frac{\delta_j L_i L_j}{1 + \delta_j L_j} \gamma_i^\top \gamma_j dt + \sum_{j=k}^{n-1} \frac{\delta_j L_i L_j}{1 + \delta_j L_j} \gamma_i^\top \gamma_j dt + \\ & + L_i \gamma_i^\top dW^{(k)}, \quad 0 \leq t \leq T_i \wedge T_k \wedge T_{n-1}, \end{aligned} \quad (1.19)$$

where $W^{(k)}$ is a Brownian motion under P_k . Note that for $k > i + 1$, the drift term in (1.19) is negative, for $k = i + 1$ equal to zero (see (1.15)), and for $k \leq i$ positive. In practical situations it makes sense to choose the bond measure P_k such that the drift term in (1.19) is as small as possible. For instance, when a claim involves only L_i , it is natural to take as pricing measure P_{i+1} . Usually, however, one needs a Libor measure which lives on the whole tenor structure like the forward bond or terminal measure P_n . But, the drift of L_i in this measure contains $n - i$ terms and so may be large when i is small. This may cause undesirable numerical errors (e.g., in an SDE simulation with time discretization steps $\Delta t_k = \delta_k$). Therefore, as an alternative, we introduce a new measure on the whole tenor structure, called the *spot Libor measure*.

The spot Libor measure

As mentioned above, we do not explicitly assume a continuous savings account

$$\beta_t := e^{\int_0^t r(s)ds},$$

hence a tradable short rate r , since this is not necessary for pricing Libor derivative claims. As an alternative we consider a discrete savings account generated by the spot Libor rate in the following way. We start at $t = T_0 = 0$ with an initial capital \$1 invested in the zero bond B_1 . At T_1 we reinvest (roll over) the account value $1/B_1(0)$ in zero bond B_2 , yielding an account value

$$\frac{1}{B_1(0)B_2(T_1)} = \frac{1}{B_1(0)}(1 + \delta_1 L_1(T_1)),$$

at T_2 and so on. We thus end up with the so called spot Libor rolling over account,

$$B_*(0) := 1, \quad B^*(t) := \frac{B_{m(t)}(t)}{B_1(0)} \prod_{i=1}^{m(t)-1} (1 + \delta_i L_i(T_i)), \quad 0 < t \leq T_n, \quad (1.20)$$

where $m(t) := \min\{m : T_m \geq t\}$ denotes the next reset date at time t . Clearly, the process B_* is a continuous and completely determined by the Libors at the tenor dates. Moreover, B_* can be seen as the value process of a self financing portfolio (φ, B) , $B^* = \varphi^\top B$, where

$$\varphi_i(t) = \frac{1_{(T_{i-1}, T_i]}(t)}{B_1(0)} \prod_{i=1}^{m(t)-1} (1 + \delta_i L_i(T_i)), \quad 1 < i \leq n, \quad 0 < t \leq T_n,$$

with $\varphi_i(0) := \varphi_i(0+)$. Hence, φ is left continuous and by Lemma 1.1 it follows that ξB_* is a martingale. So we may introduce the *spot Libor measure* P^* according to Definition 1.5 by

$$\frac{dP^*}{dP} := \frac{\xi(T_n)B^*(T_n)}{B_*(0)} = \frac{\xi(T_n)}{B_1(0)} \prod_{i=1}^{n-1} (1 + \delta_i L_i(T_i)).$$

Note that B^* is only defined on $[0, T_n]$, so here we have to replace T_∞ in Definition 1.5 by T_n .

Let us now derive the Libor dynamics in P_* . From (1.20) it follows that

$$\begin{aligned} \frac{dB_*(t)}{B_*(t)} 1_{(T_{i-1}, T_i]}(t) &= 1_{(T_{i-1}, T_i]}(t) \frac{dB_i(t)}{B_i(t)} \\ &= 1_{(T_{i-1}, T_i]}(t) (\mu_i dt + \sigma_i^\top dW), \quad 1 < i \leq n \end{aligned}$$

(in the objective measure), hence the SDE

$$\frac{dB_*(t)}{B_*(t)} = \mu_{m(t)} dt + \sigma_{m(t)}^\top dW, \quad 0 \leq t \leq T_n,$$

describes the dynamics of the spot Libor rolling-over account B_* . By straightforward application of Itô via Lemma A.2 and using (1.6) we have for any j ,

$$\begin{aligned} d \frac{B_j(t)}{B_*(t)} &= \frac{B_j(t)}{B_*(t)} ((\mu_j - \mu_{m(t)} - (\sigma_j - \sigma_{m(t)})^\top \sigma_{m(t)}) dt + (\sigma_j - \sigma_{m(t)})^\top dW) \\ &= \frac{B_j(t)}{B_*(t)} (\sigma_j - \sigma_{m(t)})^\top ((\lambda - \sigma_{m(t)}) dt + dW) \\ &=: \frac{B_j(t)}{B_*(t)} (\sigma_j - \sigma_{m(t)})^\top dW^*, \quad 0 \leq t \leq T_j. \end{aligned}$$

Due to Lemma 1.2, B_i/B_* is a P_* -martingale and so W^* is standard Brownian in P_* . Then, by (1.14) we may write

$$dW^{(n)} = dW^* + (\sigma_{m(t)} - \sigma_n) dt, \quad (1.21)$$

and by plugging (1.21) into (1.16) and using (1.13) we obtain the Libor dynamics in P_* ,

$$dL_i = \sum_{j=m(t)}^i \frac{\delta_j L_i L_j}{1 + \delta_j L_j} \gamma_i^\top \gamma_j dt + L_i \gamma_i^\top dW^*, \quad 0 \leq t \leq T_i. \quad (1.22)$$

We see that in the spot Libor measure L_i in (1.22) contains $i - m(t)$ drift terms, whereas the SDE in the terminal measure (1.16) contains $n - i$ drift terms. For numerical reasons it is important to keep the Libor drift as small as possible. Therefore, loosely speaking, for products involving only short maturity Libors the dynamics in the spot Libor measure (1.22) is preferable whereas for longer dated products the representation in the terminal measure (1.16) may be recommended.

For practical applications a very important class of Libor models consists of the so called Libor *Market* models:

DEFINITION 1.6 *A Libor model where the volatility process $t \rightarrow \gamma(t)$ is a deterministic function is called a **Libor Market Model (LMM)**.*

From (1.15) we see that in a Libor market model L_i is a *log-normal martingale* in the measure P_{i+1} for $1 \leq i < n$. This is one of the main features of a Libor market model because, as a consequence, important liquidly traded market instruments such as caps (see Section 1.3.1) can be priced with closed form (Black-Scholes type) expressions.

1.2.2 Swap Rate Processes and Measures

An interest rate payer swap on a loan over a period $[T_p, T_q]$, with notional amount \$1, is a contract to pay a fixed rate κ and to receive spot Libor (floating) at the settlement dates $T_{p+1}, \dots, T_q$. By a simple portfolio argument we can show that the arbitrage-free value of this contract at $t \leq T_p$ is equal to

$$Swap_{p,q;\kappa}(t) := B_p(t) - B_q(t) - \sum_{j=p}^{q-1} \kappa \delta_j B_{j+1}. \quad (1.23)$$

Indeed, on the one hand, the fixed interest rate payments $\kappa \delta_j$ at the dates $T_{p+1}, \dots, T_q$ have at time t the value of $\sum_{j=p}^{q-1} \kappa \delta_j B_{j+1}$. On the other hand, the future floating payments to be received are at time t equivalent to a portfolio consisting of a zero bond B_p *long* and B_q *short*, hence $B_p - B_q$. As a result, the difference (1.23) is the value of the swap contract at time t .

The *swap rate* $S_{p,q}$ is defined as *that* fixed rate κ for which in (1.23) $Swap_{p,q;\kappa} = 0$. Hence,

$$S_{p,q}(t) = \frac{B_p(t) - B_q(t)}{\sum_{j=p}^{q-1} \delta_j B_{j+1}(t)} =: \frac{B_p(t) - B_q(t)}{B_{p,q}(t)}, \quad (1.24)$$

where $B_{p,q}$ is the so called annuity numeraire.

We refer to (1.24) as the standard swap rate. In practice, however, a swap contract may be defined such that fixed rates need to be settled only at a real subset of the Libor tenor dates in the respective swap period. For example, in the US, UK, and Japanese market Libors are quarterly, whereas swaps are annually settled. In the Euro market Libors are semi-annual, whereas swaps are annually settled. In order to cover these situations we consider a generalized swap contract over a period $[T_p, T_q]$, for a fixed choice of p and q , where the settlement dates are given by a sequence $(T_{p_1}, \dots, T_{p_k})$ with $T_p =: T_{p_0} < T_{p_1} < \dots < T_{p_k} = T_q$. This leads to the corresponding generalized swap contract

$$Swap_{p,q;\kappa}^{\text{gen}}(t) := B_p(t) - B_q(t) - \sum_{j=0}^{k-1} \kappa (T_{p_{j+1}} - T_{p_j}) B_{p_{j+1}}(t), \quad (1.25)$$

and generalized swap rate

$$S_{p,q}^{\text{gen}}(t) = \frac{B_p(t) - B_q(t)}{\sum_{j=0}^{k-1} (T_{p_{j+1}} - T_{p_j}) B_{p_{j+1}}(t)} =: \frac{B_p(t) - B_q(t)}{B_{p,q}^{\text{gen}}(t)}, \quad (1.26)$$

with $B_{p,q}^{\text{gen}}$ being the generalized annuity numeraire. For example, taking $p_j = p + j$, $j = 0, \dots, k := q - p$, yields the standard swap rate again. Henceforth we drop the superscript (gen) since it will be always clear from the context which kind of swap rate is under consideration.

Let us consider the dynamics of the (generalized) annuity numeraire in (1.26). We may write

$$\begin{aligned} dB_{p,q} &= \sum_{j=0}^{k-1} (T_{p_{j+1}} - T_{p_j}) dB_{p_{j+1}} \\ &= B_{p,q} \sum_{j=0}^{k-1} \frac{(T_{p_{j+1}} - T_{p_j}) B_{p_{j+1}}}{B_{p,q}} (\mu_{p_{j+1}} dt + \sigma_{p_{j+1}}^\top dW) \\ &= B_{p,q} \left(\sum_{j=0}^{k-1} w_j^{p,q} \mu_{p_{j+1}} dt + \sum_{j=0}^{k-1} w_j^{p,q} \sigma_{p_{j+1}}^\top dW \right), \end{aligned}$$

with $w_j^{p,q} = \frac{(T_{p_{j+1}} - T_{p_j}) B_{p_{j+1}}}{B_{p,q}}$. By using Itô via Lemma A.2 we then obtain for any $p \leq r \leq q$,

$$\begin{aligned} \frac{d(B_r/B_{p,q})}{(B_r/B_{p,q})} &= (\mu_r - \sum_{j=0}^{k-1} w_j^{p,q} \mu_{p_{j+1}} - \sum_{l=0}^{k-1} w_l^{p,q} \sigma_{p_{l+1}}^\top (\sigma_r - \sum_{j=0}^{k-1} w_j^{p,q} \sigma_{p_{j+1}})) dt \\ &\quad + (\sigma_r - \sum_{j=0}^{k-1} w_j^{p,q} \sigma_{p_{j+1}})^\top dW. \end{aligned}$$

Since the $w_j^{p,q}$ are weights, i.e., $\sum_{j=0}^{k-1} w_j^{p,q} = 1$, we get by using (1.6),

$$\begin{aligned} \frac{d(B_r/B_{p,q})}{(B_r/B_{p,q})} &= \sum_{j=0}^{k-1} w_j^{p,q} (\sigma_r - \sigma_{p_{j+1}})^\top \left(\lambda dt - \sum_{l=0}^{k-1} w_l^{p,q} \sigma_{p_{l+1}} dt + dW \right) \\ &=: \sum_{j=0}^{k-1} w_j^{p,q} (\sigma_r - \sigma_{p_{j+1}})^\top dW^{p,q}. \end{aligned} \quad (1.27)$$

Both ξB_r and $\xi B_{p,q}$ are martingales, so $W^{p,q}$ is standard Brownian motion in the measure $P_{p,q}$, induced by the annuity numeraire $B_{p,q}$ according to Lemma 1.2. By now applying (1.27) for $r = p$ and $r = q$ to the swap rate

defining expression (1.26), respectively, we obtain

$$\begin{aligned}
 dS_{p,q} &= \left(\frac{B_p}{B_{p,q}} \sum_{j=0}^{k-1} w_j^{p,q} (\sigma_p - \sigma_{p_{j+1}}) - \frac{B_q}{B_{p,q}} \sum_{j=0}^{k-1} w_j^{p,q} (\sigma_q - \sigma_{p_{j+1}}) \right)^\top dW^{p,q} \\
 &= S_{p,q} \left(\sum_{j=0}^{k-1} w_j^{p,q} (\sigma_p - \sigma_{p_{j+1}}) + \frac{B_q}{B_p - B_q} (\sigma_p - \sigma_q) \right)^\top dW^{p,q} \\
 &=: S_{p,q} \sigma_{p,q}^\top dW^{p,q},
 \end{aligned} \tag{1.28}$$

where $\sigma_{p,q}$ is the relative volatility process of the swap rate $S_{p,q}$.

Analogue to the definition of a Libor market model we now introduce the notion of a swap market model.

DEFINITION 1.7 *If in (1.28) the volatility process $t \rightarrow \sigma_{p,q}(t)$ is a deterministic function we speak of a **Swap Market Model** for $S_{p,q}$.*

In a swap market model $\sigma_{p,q}$ is a *log-normal martingale* in the measure $P_{p,q}$. As an important consequence, European options on swap contracts over $[T_p, T_q]$, called *swaptions*, can be priced exactly with Black-Scholes type formulas; see Section 1.3.2. Moreover, as we will show, there exist very accurate swaption approximation formulas for swaptions in a Libor market model (see Section 1.3.3).

By using (1.13) we may express $\sigma_{p,q}$ in (1.28) in the Libor volatility structure to yield

$$\begin{aligned}
 \sigma_{p,q} &= \sum_{j=0}^{k-1} w_j^{p,q} \sum_{l=p}^{p_{j+1}-1} \frac{\delta_l L_l}{1 + \delta_l L_l} \gamma_l + \frac{B_q}{B_p - B_q} \sum_{l=p}^{q-1} \frac{\delta_l L_l}{1 + \delta_l L_l} \gamma_l \\
 &= \sum_{l=p}^{q-1} \gamma_l \frac{\delta_l L_l}{1 + \delta_l L_l} \left(\sum_{j=j_l}^{k-1} w_j^{p,q} + \frac{B_q}{B_p - B_q} \right)
 \end{aligned} \tag{1.29}$$

with $j_l := \min\{j : p_{j+1} \geq l + 1\}$. Expression (1.29) will be the starting point for deriving swaption approximation formulas in a Libor market model in Section 1.3.3.

Libor and swap market models are in fact inconsistent with each other. For example, if L_1 and L_2 have deterministic volatilities it follows immediately from (1.29) that the volatility of the standard swap $S_{1,3}$ is non-deterministic. Therefore one usually chooses that market model which is most appropriate for pricing and hedging a particular product under consideration.

REMARK 1.3 In a Libor market model where $n - 1$ Libor volatilities γ_i are specified by deterministic vector functions, the bond volatility differences

$\sigma_i - \sigma_k$ are implicitly determined via (1.13). Hence the bond volatilities are determined up to one degree of freedom. In contrast, for swap market models in general one cannot specify simultaneously $n(n-1)/2$ deterministic swaption volatilities $\sigma_{p,q}$ for all $1 \leq p < q \leq n$, since regarding (1.28) we have only n bond volatilities to determine them, hence n degrees of freedom. But, for instance, for a complete system of standard swaps it is possible to choose $\sigma_{1,n}, \dots, \sigma_{n-1,n}$ simultaneously deterministic. $\square$

1.3 Pricing of Caps and Swaptions in Libor and Swap Market Models

1.3.1 Libor Caps and Caplets

A Libor cap on a loan over period $[T_p, T_q]$, with notional amount \$1, is a contract to pay at the settlement dates $T_{p+1}, \dots, T_q$, floating spot Libor capped with a pre-specified strike rate κ . Clearly, this contract is equivalent to a sequence of cash-flows $(L_j(T_j) - \kappa)^+ \delta_j$, $p \leq j < q$, at the respective settlement dates. Every T_{j+1} pay-off is homogeneous of degree 1, since it can be written as $B_{j+1}(T_{j+1})(L_j(T_j) - \kappa)^+ \delta_j$. So, under the measurability conditions in Theorem 1.3 (which are trivially satisfied for Libor volatility structures of type $\gamma(t, L)$), the value of this contract at $t \leq T_p$ is given by

$$Cap_{p,q;\kappa}(t) := \sum_{j=p}^{q-1} B_{j+1}(t) E_{j+1}^{\mathcal{J}_t} (L_j(T_j) - \kappa)^+ \delta_j, \quad (1.30)$$

where $(\mathcal{J}_t)_{0 \leq t \leq T_{n-1}}$ is the filtration generated by the Libor process. In (1.30) each T_{j+1} cash-flow can be seen as the cash-flow of a “mini cap” over T_j, T_{j+1} , called a *caplet*, and is priced in its canonical measure P_{j+1} via Proposition 1.2. In a Libor market model caplets can be priced with Black-Scholes type expressions and so, by (1.30), caps also.

PROPOSITION 1.4

The price of a caplet over $[T_j, T_{j+1}]$ with strike κ in a LMM with deterministic volatility structure $t \rightarrow \gamma(t)$ is given by

$$\begin{aligned} Caplet_{j,j+1;\kappa}(t) &= B_{j+1}(t) E_{j+1}^{\mathcal{J}_t} (L_j(T_j) - \kappa)^+ \delta_j \\ &= B_{j+1}(t) (\delta_j L_j(t) \mathcal{N}(d_+) - \delta_j \kappa \mathcal{N}(d_-)), \end{aligned} \quad (1.31)$$

with $\mathcal{N}$ being the standard normal distribution and

$$d_{\pm} = \frac{\ln(L_j(t)/\kappa) \pm (\sigma_j^B)^2 (T_j - t)/2}{\sigma_j^B \sqrt{T_j - t}}, \quad (\sigma_j^B)^2 = \frac{1}{T_j - t} \int_t^{T_j} |\gamma_j|^2(s) ds.$$

PROOF Since γ_i is deterministic, L_j is a log-normal martingale in P_{j+1} by (1.15). In particular we have

$$\begin{aligned} L_j(T_j) &= L_j(t) \exp \left(-\frac{1}{2} \int_t^{T_j} |\gamma_j|^2(s) ds + \int_t^{T_j} \gamma_j(s)^\top dW^{(j+1)}(s) \right) \\ &\stackrel{dist.}{=} L_j(t) \exp \left(-\frac{1}{2} \int_t^{T_j} |\gamma_j|^2(s) ds + \sigma_j^B \sqrt{T_j - t} \zeta \right), \end{aligned}$$

for a real valued standard normally distributed ζ . Now the statement follows from (1.31) by elementary calculus. $\square$

Formula (1.31), essentially due to Black (1976), is usually called the Black (76) formula.

1.3.2 Swaptions in a Swap Market Model

A European (payer) swaption over period $a [T_p, T_q]$ gives the right to enter at T_p into an interest rate swap with strike rate κ , as introduced in Section 1.2.2. Clearly, the swaption contract is equivalent to a (degree 1 homogeneous) cash-flow at T_p which is equal to $B_p(T_p) (Swap_{p,q;\kappa}(T_p))^+$. Hence the swaption value at time $t \leq T_p$ is given by

$$\begin{aligned} Swpn_{p,q;\kappa}(t) &= B_{p,q}(t) E_{p,q}^{\mathcal{J}_t} \frac{(Swap_{p,q;\kappa}(T_p))^+}{B_{p,q}(T_p)} \\ &= B_{p,q}(t) E_{p,q}^{\mathcal{J}_t} (S_{p,q}(T_p) - \kappa)^+; \end{aligned} \quad (1.32)$$

see (1.23) and (1.25).

In a swap market model the swap rate volatility is deterministic. As a consequence, in a SMM the swap rate $S_{p,q}$ is a log-normal martingale and, similar to Proposition 1.4, swaptions can be priced analytically by Black-Scholes type formulas. Indeed, if $t \rightarrow \sigma_{p,q}$ is deterministic, we may write by (1.28) for $t \leq T_p$,

$$S_{p,q}(T_p) = S_{p,q}(t) \exp \left(-\frac{1}{2} \int_t^{T_p} |\sigma_{p,q}|^2(s) ds + \int_t^{T_p} \sigma_{p,q}(s)^\top dW^{p,q}(s) \right)$$

and obtain from (1.32) the so called Black swaption formula,

$$Swpn_{p,q;\kappa}(t) = B_{p,q}(t) (S_{p,q}(t) \mathcal{N}(d_+) - \kappa \mathcal{N}(d_-)), \quad \text{with} \quad (1.33)$$

$$d_{\pm} := \frac{\ln(S_{p,q}(t)/K) \pm (\sigma_{p,q}^B)^2(T_p - t)/2}{\sigma_{p,q}^B \sqrt{T_p - t}}, \quad (\sigma_{p,q}^B)^2 = \frac{1}{T_p - t} \int_t^{T_p} |\sigma_{p,q}|^2(s) ds.$$

1.3.3 Approximating Swaptions in a Libor Market Model

The dynamics of swap rates in a Libor market model is rather complicated due to the stochastic factors behind the γ_l in (1.29). Therefore, closed form pricing of swaptions in a Libor market model is in general not possible. But, it is nonetheless possible to give surprisingly accurate swaption approximation formulas in an LMM.

Using (1.29) and a little algebra we may express for fixed p and q the squared magnitude of the swap volatility by

$$|\sigma_{p,q}|^2 = \frac{1}{S_{p,q}^2} \sum_{l=p}^{q-1} \sum_{l'=p}^{q-1} \gamma_l^\top \gamma_{l'} L_l L_{l'} v_l^{p,q} v_{l'}^{p,q}, \quad \text{with} \quad (1.34)$$

$$v_l^{p,q} = \frac{\delta_l}{1 + \delta_l L_l} \left(\sum_{j=j_l}^{k-1} w_j^{p,q} \frac{B_p - B_q}{B_{p,q}} + \frac{B_q}{B_{p,q}} \right), \quad j_l = \min\{j : p_{j+1} \geq l + 1\}. \quad (1.35)$$

Let us write the possibly non-standard swap rate $S_{p,q}(t)$, $t \leq T_p$, as a sum of forward Libors,

$$S_{p,q}(t) = \frac{B_p(t) - B_q(t)}{B_{p,q}(t)} = \sum_{j=p}^{q-1} \frac{B_j(t) - B_{j+1}(t)}{B_{p,q}(t)} = \sum_{j=p}^{q-1} \hat{v}_j^{p,q}(t) L_j(t), \quad (1.36)$$

where $B_{p,q}$ is the annuity numeraire in (1.24) or (1.26) and $\hat{v}_j^{p,q} = \delta_j B_{j+1} / B_{p,q}$. In the case of a standard swap we have $\sum_{j=p}^{q-1} \hat{v}_j^{p,q} = 1$, so then the coefficients $v_j^{p,q}$ can be seen as a weights. We now derive a relation between $\hat{v}$ in (1.36) and v in (1.34). Consider the system of (virtual) swap rates $S_{l,q}^0$, for $l = p, \dots, q-1$, defined by

$$S_{l,q}^0 = \frac{B_l - B_q}{B_{l,q}}, \quad \text{with} \quad (1.37)$$

$$B_{l,q} := \sum_{j=j_l}^{k-1} (T_{p_{j+1}} - T_{p_j}) B_{p_{j+1}} = B_{p,q} \sum_{j=j_l}^{k-1} w_j^{p,q}.$$

Note that $S_{l,q}^0$ can be seen as the swap rate over period $[T_l, T_q]$ with settlement dates $T_{j_l}, \dots, T_{p_k} = T_q$. By (1.35) and (1.37) it easily follows that

$$\begin{aligned} v_l^{p,q} - \hat{v}_l^{p,q} &= \frac{\delta_l}{1 + \delta_l L_l} \left(\sum_{j=j_l}^{k-1} w_j^{p,q} \frac{B_p - B_q}{B_{p,q}} + \frac{B_q}{B_{p,q}} \right) - \frac{\delta_l B_{l+1}}{B_{p,q}} \\ &= \frac{\delta_l B_{l,q}}{B_{p,q}} \frac{S_{p,q}^0 - S_{l,q}^0}{1 + \delta_l L_l} =: \hat{y}_l^{p,q}, \quad p \leq l < q. \end{aligned} \quad (1.38)$$

The $\hat{y}$ terms in (1.38) have magnitudes comparable with differences of swap rates; hence they are usually rather small. Moreover we have $\hat{y}_l^{p,q} = 0$ when

$S_{l,q}^0 = S_{p,q}$ for $p < l < q$. For example, this is the case for standard swaptions when the yield curve is flat (see also Jäckel & Rebonato 2003).

Let us integrate (1.34) to obtain

$$\frac{1}{T_p - t} \int_t^{T_p} |\sigma_{p,q}|^2(s) ds = \sum_{l,l'=p}^{q-1} \int_t^{T_p} \frac{v_l^{p,q}(s) v_{l'}^{p,q}(s) L_l(s) L_{l'}(s)}{S_{p,q}^2(s)} \gamma_l(s)^\top \gamma_{l'}(s) ds, \quad (1.39)$$

where $t \leq T_p$ denotes the present calendar date. We now note that the (stochastic) fractions in the r.h.s. integrands of (1.39), which by (1.36), (1.38) add up to approximately one and thus may be regarded as weights, tend to vary relatively slow in practice and therefore may be approximated by their values at t . Under this additional assumption instantaneous swap volatilities may be considered as deterministic (though model inconsistent). As a result the quantities in the l.h.s. of (1.39) may be seen as squares of implied Black volatilities of approximative swaption prices in the Libor model and we thus obtain for a swaption with unit nominal and strike κ the following approximation formula.

General swaption approximation formula

$$\begin{aligned} \text{Swpn}_{p,q;\kappa}(t) &\approx B_{p,q}(t) (S_{p,q}(t) \mathcal{N}(d_+) - \kappa \mathcal{N}(d_-)), \\ d_{\pm} &= \frac{\ln(S_{p,q}(t)/\kappa) \pm (\sigma_{p,q}^B)^2 (T_p - t)/2}{\sigma_{p,q}^B \sqrt{T_p - t}}, \text{ where} \\ (\sigma_{p,q}^B)^2 &= \sum_{l,l'=p}^{q-1} \frac{v_l^{p,q}(t) v_{l'}^{p,q}(t) L_l(t) L_{l'}(t)}{S_{p,q}^2(t)} \int_t^{T_p} \gamma_l(s)^\top \gamma_{l'}(s) ds, \\ \text{with } v_l^{p,q} &= \frac{\delta_l B_{l+1}}{B_{p,q}} + \hat{y}_l^{p,q}. \end{aligned} \quad (1.40)$$

The $\hat{y}$ -term can be computed by (1.38). By the formula (1.40) we can evaluate quasi-analytically approximative model swaption prices which should be computed otherwise by tedious Monte Carlo simulation. This is very important for calibration procedures where the market model is calibrated by fitting the volatilities (1.40) directly to At-The-Money swaption volatilities quoted in the market.¹

For our applications we highlight the following versions of the general swaption approximation formula (1.40).

• Simple swaption approximation

$$\hat{y}_l^{p,q} \equiv 0 \quad (1.41)$$

¹For ATM instruments calibrating to volatilities or to prices makes in practice hardly any difference.

in (1.40) has the following important features.

- For a flat yield curve and equally settled Libors and swaps this approximation coincides with the general approximation (1.40).
- In the simple approximation the Black volatilities in (1.40) are the same for all swap settlement conventions, since in the expression for the Black volatility in (1.40) the annuity numeraire $B_{p,q}$ cancels out if $\hat{y} = 0$. Note, however, swap prices and the quality of approximation are generally not the same for all swap conventions.

• **Refined swaption approximation in different markets**

Let m be an integer, p, q such that $q - p \bmod m = 0$, and set $p_j := p + mj$. It then follows that $j_l = [(l-p)/m]$, with $[x] := \max\{k \in \mathbb{Z} : k \leq x\}$ denoting the Entier function, and obtain for the correction term (1.38) by a little algebra,

$$\hat{y}_l^{p,q} = \frac{\delta_l}{B_{p,q}(1 + \delta_l L_l)} \left(B_p - B_l - S_{p,q} \sum_{j=0}^{[\frac{l-p}{m}]-1} B_{p+m(j+1)} \sum_{r=p}^{p+m-1} \delta_{mj+r} \right). \quad (1.42)$$

Thus, (1.42) yields refined swaption approximation formulas for application in different markets:

- $m = 1$ for the standard (equally settled) swaption market,
- $m = 2$ for Euro swaptions in a semi-annual Libor model,
- $m = 4$ for US, UK, and Japanese swaptions in a quarterly Libor model.

The fact that for standard swaptions the simple swaption approximation (1.40), (1.41) and the refined approximation (1.40), (1.42) coincide when the initial yield curve is flat was pointed out and explained previously by Jäckel & Rebonato (2003). For a typical flat yield curve and LMM volatility structure the relative errors between approximated swaption prices due to (1.40), (1.41) and Monte Carlo simulated prices turned out to be about 0.3%. For a non-flat yield curve and refined formula (1.40), (1.42) with $m = 1$, the relative errors turned out to be 0.3% as well relative in this paper.

The conclusions of Jäckel & Rebonato (2003) were confirmed by our simulation tests. But, we then considered annually settled swaptions in a semi-annual Libor model, for instance, for modelling an EurIbor environment. By comparing (accurate enough) Monte Carlo prices with approximations due to (1.40), (1.41) for a flat yield curve, our experiments showed that the accuracy of the simple approximation (1.40), (1.41) was significantly less, namely, about 1% relative for a flat yield curve and about 2% relative for some typical non-flat curve. Of course this is due to the fact that the $\hat{y}$ term in (1.40) not vanishes for non-standard swaptions in general. However, by applying the

refined formula (1.40), (1.42) with $m = 2$ we obtained relative errors of about 0.3% again. We therefore draw the following conclusion.

For differently settled caps and swap(tion)s, in particular in the Euro, US, UK, and Japanese market, we recommend to use the refined swaption formula (1.40), (1.42) for a suitable m in any case, whether the initial yield curve is flat or not.

1.3.4 Smile/Skew Extensions of the Libor Market Model

We have seen that a caplet can be priced by Black's formula (1.31) in a LMM. Conversely, given a caplet price, strike, and maturity, one can ask for the so called *implied caplet volatility* such that (1.31) holds. In the markets, caps, caplets, and swaptions are usually quoted by (implied) volatilities rather than prices. Obviously, the implied Black volatility of a caplet price due to a Libor market model is flat with respect to caplet strike and maturity. This is actually not consistent with reality, where usually caplet volatilities as function of the strike behave like a skew, or an asymmetric smile.

An obvious way to allow for smile effects in the Libor model is to consider in (1.22) instead of a deterministic volatility structure a more general volatility process of the form, $t \rightarrow \gamma_i(t, L(t))$, where for $1 \leq i < n$, $\gamma_i(\cdot, \cdot)$ is a certain deterministic $\mathbb{R}^m$ -valued vector function which satisfies suitable conditions. If we take $\gamma_i(t, L) = \gamma_i^{(0)}(t)L_i^{\alpha-1}$, where $\alpha > 0$ and $t \rightarrow \gamma_i^{(0)}(t) \in \mathbb{R}^m$ is deterministic, we have the so called CEV model which allows for closed form caplet prices. For details see Schroder (1989) and Andersen & Andreasen (2000), where a finite difference procedure for a volatility structure of the form $\gamma_i(t, L) = \gamma^{(0)}(t)\phi(L_i)$, $1 \leq i < n$, is proposed. As a more general approach to cover smile effects in model prices of caps and swaptions, one considers in the literature Libor models driven by Lévy processes rather than Brownian motions; see Eberlein & Özkan (2002), Glasserman & Kou (2003), Glasserman & Merener (2002), and Jamshidian (1999).

While extensions of the Libor market model give more flexibility and thus may reflect reality more properly, the pricing of caps and swaptions is generally more complicated and time consuming in such models. This may be a problem when these pricing procedures are used in a procedure of calibrating to caps and swaptions. Further, in particular for jump-diffusion Libor models, the larger model flexibility may lead to serious instability of calibration procedures. For a treatment of stability problems connected with the identification of Lévy based models, such as jump-diffusions, we refer to Cont & Tankov (2003). In this book we restrict ourselves to the standard Libor market model and its calibration issue, although we do not want to say that more general Libor models are superfluous luxury. For instance, it will be certainly possible to extend the calibration procedures for a parametric Libor model proposed in Chapter 3 to certain generalized Libor models, for example the CEV model.